

Algorithmic Trading of Bitcoin Using Wikipedia and Google Search Volume

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Abstract

Trading and speculating on Bitcoin is difficult. With large daily price movements and frequent price crashes, implementing quantitative trading strategies can be challenging. This paper presents a simple trading algorithm which uses the volume of Wikipedia and Google Trends search terms to generate trading signals for buying or selling Bitcoin. By exploiting the collective interest in Bitcoin as measured by search volume, findings suggest that highly profitable trading strategies can be constructed that avoid market crashes. The strategy based on the “Bitcoin” Google and Wikipedia search terms generates a cumulative return of 9418.04% and 3549.05% respectively.

1 Introduction

Over the last year interest in cryptocurrencies has reached unprecedented highs. The price of Bitcoin (BTC) has grown exponentially, with many economists and market commentators calling it the largest speculative bubble the world has seen. Whether cryptocurrencies in general hold any fundamental value is a topic of philosophical debate; currently, price is primarily determined largely by speculative trade rather than any intrinsic or fundamental value. With annualized volatility often exceeding 100%, BTC presents traders with opportunity to profit using highly advanced trading algorithms. The introduction of blockchain technology and cryptocurrency as an asset class has seen a rapid uptake in interest over the last year – primarily due to the aforementioned increase in the price of BTC. Bitcoin was created by an anonymous person or organisation known as Satoshi Nakamoto (Nakamoto, 2008). Studies investigating the feasibility of a trading strategy based on Wikipedia and Google search volume, to the author’s knowledge, have yet to be published. Work on quantifying trading behaviour in stock markets using Google Trends search volume, however, has been well documented by Preis et al. (2013). The study shows that Google Trends offers a unique insight into the state of the economy. By using certain search terms such as ‘debt’, equity

traders can use Google Trends search volume to exploit warning signs and make substantial trading profits by using the Dow Jones Industrial Index (DJIA) as the investment proxy. The study concludes that, besides being able to make profit by exploiting search volume data, combining large behavioural data sets (such as financial trading data) with Google search volumes, offers unique insights into the collective decision-making process of the market.

The correlation of BTC price with Google and Wikipedia search term volumes was first documented by Kristoufek (2013). Apart from a very strong, statistically significant correlation coefficient, the study found that there is a bidirectional, causal relationship between BTC price and the “Bitcoin” Google search term volume. Furthermore, when prices are above the moving average search volume, increasing interest (measured by search volume) pushes price upward. Findings also indicate that large price increases precede spikes in search volume. By using a Vector Autoregressive (VAR) model with one lag applied to the logarithmic differences of Bitcoin Google search volume and Bitcoin price, Kristoufek (2013) finds that a 10% change in “Bitcoin” search volume yields a price increase of 2% over the following two weeks. Kristoufek’s hypothesis indicates that a rising interest in BTC measured by search volume puts pressure on demand, raising the price which causes more people to gain interest (including the general public). This creates a cycle, which Kristoufek believes is the cause of BTC price bubbles and subsequent crashes.

Garcia & Schweitzer (2015) use sentiment analysis based on tweets about BTC and cryptocurrency to derive profitable trading strategies. Their findings indicate an increase in the polarization of opinion and exchange volume precedes rising BTC prices. Both studies show that big data sources are useful predictors of BTC price movements.

This paper aims to investigate whether a very simple trading algorithm can be constructed which generates buy (long) and sell (short) signals from Google and, more interestingly, Wikipedia search volumes to create a BTC trading strategy that both outperforms a buy and hold strategy, with lower risk. The primary motivation for the strategy is the extreme volatility of BTC prices and resulting risk of large losses. The prior findings by Preis et al. (2013), Bollen et al. (2013) and others suggest the collective decision-making processes of traders and speculators are captured by ‘big data’ sources such as Google or Twitter. It would be of great benefit to BTC traders and new investors if Wikipedia and Google search volume could give predictive insight to market sentiment. Not only could the market be timed but large crashes could be avoided by basing investing decisions on keyword search volumes.

2 Methods & Results

Due to the remarkably high volatility of BTC prices, all trading signals are generated on a daily basis. Six keywords are used; three are directly related to cryptocurrencies and three which did not have a discernible trend or pattern. The three words not related to cryptocurrency are used purely to test whether the returns achieved using cryptocurrency related search volumes are not due to chance or luck. Advanced analysis could have been used to choose ‘benchmark’ keywords – but this is beyond the scope of this study. The cryptocurrency related keywords include “Bitcoin”, “Blockchain” and “Cryptocurrency”. The benchmark words include “Happiness”, “Psychology” and “Big Data”. Keywords from Google Trends are prefixed with a ‘g’ while Wikipedia keywords are prefixed with a ‘w’. Data for daily Wikipedia search volume was obtained from tools.wmflabs.org/pageviews. Data for weekly Google search volume was obtained from trends.google.com/trends. Daily BTC price data was downloaded from coinmarketcap.com. All relevant data was accessed and downloaded on the 4th March 2018.

Daily data for Wikipedia is only available from the 1st July 2015; due to this constraint all keyword strategies are tested over the period 1st July 2015 – 3rd March 2018. Google Trends only provides data at the daily interval for periods less than approximately 90 days. Consecutive 90-day intervals can be requested, but each interval will be normalised and randomly sampled by Google. This means that each 90-day segment of the time series will not be consistent when combined. If longer periods of data are requested – Google Trends will provide a consistent time series at the weekly time interval which is not sufficient for a daily trading algorithm.

In order to address this, a multi-stage rolling window was employed which downloaded Google keyword search volume in recurring 90-day pieces with overlapping periods. Formally, from the start of the time series t , 90 (n) days of keyword data is downloaded from $[0, n]$. The next piece of data is downloaded from time $[n - 10, n + 80]$. The scaling factor is computed by dividing the last ten days of the first time series with the first ten days of the second time series (the overlapping days). The mean is then computed to yield the scaling factor which is multiplied by each element in the second time series. The two time series are now on a similar scale and can be combined. New 90-day periods are then downloaded that overlaps with the combined series until daily Google Trend search volume data is obtained for the entire period under analysis. For each keyword, 948 days of data was obtained.

Daily BTC price returns were calculated as,

$$R_t = \frac{P_t}{P_{t-1}} - 1 \quad (1)$$

2.1 Correlation Analysis

The correlation analysis is undertaken between the Wikipedia/Google search volumes and the price of BTC. Due to the mix of ordinal and ratio scaled data, the Spearman Rank correlation coefficient is used. The correlation matrix is presented in Figure 1 below.

Figure 1: Spearman Rank correlation coefficients for Google and Wikipedia search term volumes and the price of Bitcoin. It is investigated here whether there is any correlation between Google and Wikipedia search queries and the price of Bitcoin (BTC). Findings suggest that the price of BTC is strongly correlated with the cryptocurrency related key words. The correlation coefficient between BTC price and both the Wikipedia and Google search queries are all above 0.85 indicating a very strong linear relationship. The correlations between BTC price and the non-cryptocurrency related search queries are low and, in some cases, negative. Now that a statistically significant correlation between Bitcoin price movement and search query volume has been established, we can turn to developing a profitable trading strategy.

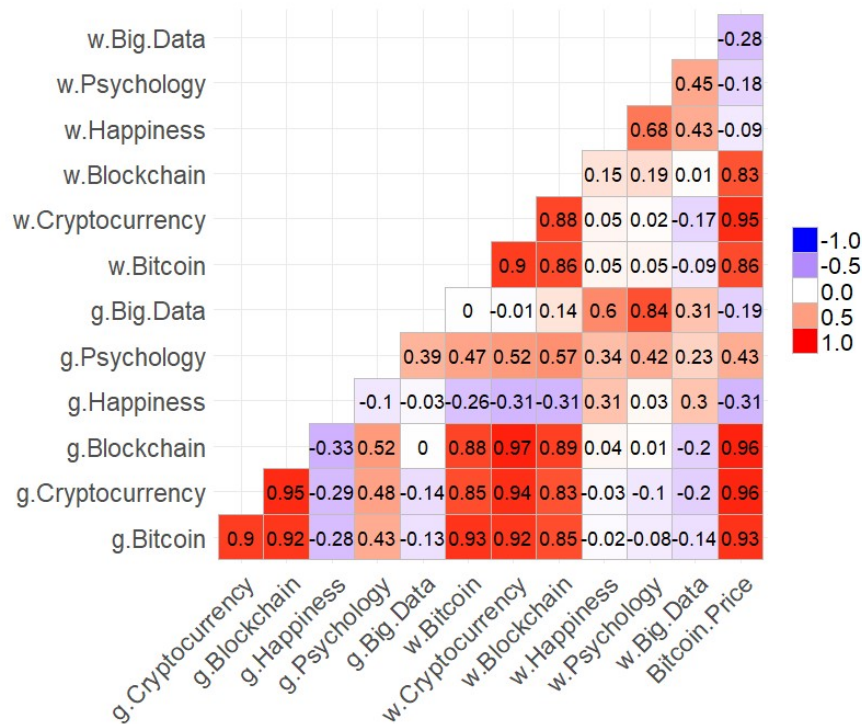
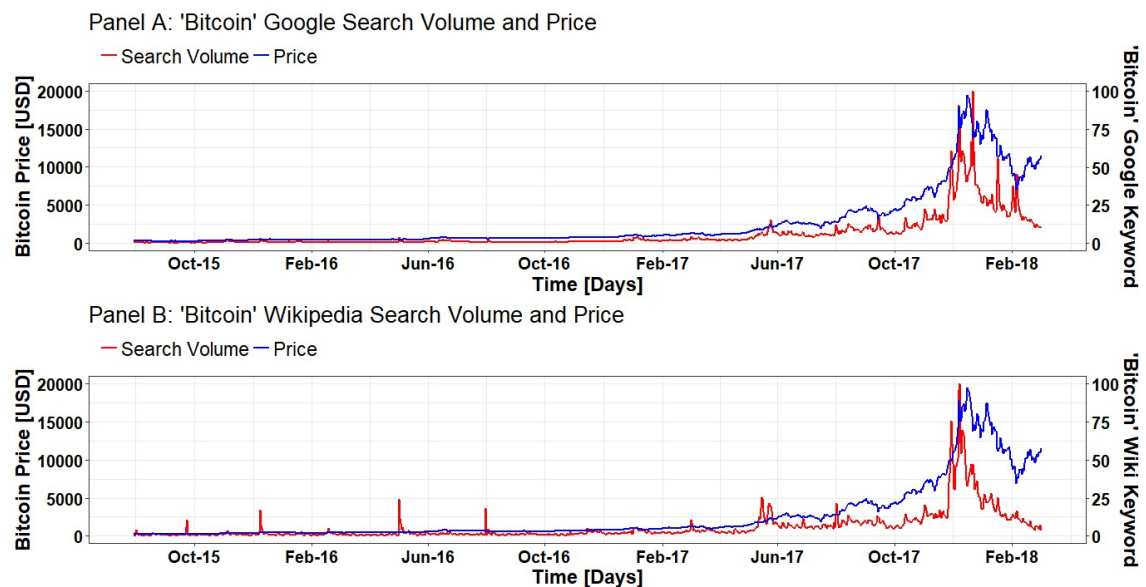


Table 1: p-Values for correlation coefficients between the price of Bitcoin and search terms.

g.Bitcoin	g.Cryptocurrency	g.Blockchain	g.Happiness	g.Psychology	g.Big.Data	N
0.00***	0.00***	0.00***	0.722	0.00***	0.00***	948
w.Bitcoin	w.Cryptocurrency	w.Blockchain	w.Happiness	w.Psychology	w.Big.Data	N
0.00***	0.00***	0.00***	0.00***	0.613	0.00***	948
Significance Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1						

Figure 2 below displays rescaled keyword data for g.Bitcoin and w.Bitcoin between 0 – 100 against the price of BTC. The relationship between the respective keywords and price is clear – the interest in BTC, measured by search volume closely follows price movement.

Figure 2: Scaled "Bitcoin" search volume from Google (Panel A) & Wikipedia "Bitcoin" search volume (Panel B). The left axis represents the BTC price in USD (the blue line), the right axis represents Google and Wikipedia search volumes scaled between 0 and 100 (the red line). It is clear to see that both Google and Wikipedia search volume closely track the price of Bitcoin. Towards the end of the graph we see a clear spike in normalised search volume which occurs just after the spike in BTC price. This shows us that the search volume is slightly lagged behind BTC price movement, i.e. a spike in BTC price will precede a spike in search volume. As the price of BTC begins to decrease, we see a marked decrease in search volume. These findings indicate that a profitable trading strategy may be able to be implemented because we will be able to buy as search volume increases and short sell when search volume declines.



2.2 Trading Strategy

The trading strategy employed is constant for all keywords under consideration. Due to the large and erratic volatility of the keyword search volume, a lagged rolling window of 28 days (4 weeks) is used to compute the median of both Wikipedia and Google Trends search volumes. In unreported results, shorter time periods created erratic trading signals that were subject to noise. The median was chosen to subvert the effect of outliers in the search volume data.

Trading signals are generated through the change in the rolling 28 day median for each trading day. A buy (long) signal is generated by a positive change in the median and a short position generated by negative change. The signals are lagged to avoid look-ahead bias. For example, if the signal generated at day $[t - 1]$ is a long position, the next days return is computed as follows,

$$StrategyReturn_t = signal_{t-1} \times R_t, \quad (2)$$

where, R_t is BTC price return for day t and $signal_{t-1}$ is a 1 (long) or -1 (short) based on the change in search volume.

Returns for the random strategy were bootstrapped by generating a time series of 10,000 random long and short positions which were then used to calculate the daily random strategy return series. The mean return for each month t was then computed. The buy and hold strategy assumes a constant long position for each trading day. Trading costs on all major cryptocurrency exchanges are minimal (less than 0.01% per trade); their effect has been excluded from the analysis.

Cumulative returns were computed by geometrically linking all daily returns over the time period covered,

$$R_{cum} = \prod_{t=1}^T (1 + R_t) \quad (3)$$

For all search term strategies, the annualized return, annualized standard deviation, annualized Sharpe ratio, Value-at-Risk (VaR) and Expected Tail Loss (ETL) were computed and are presented in Table 2 below including the cumulative return.

Table 2: Table of risk & return metrics for Wikipedia and Google keyword strategies.

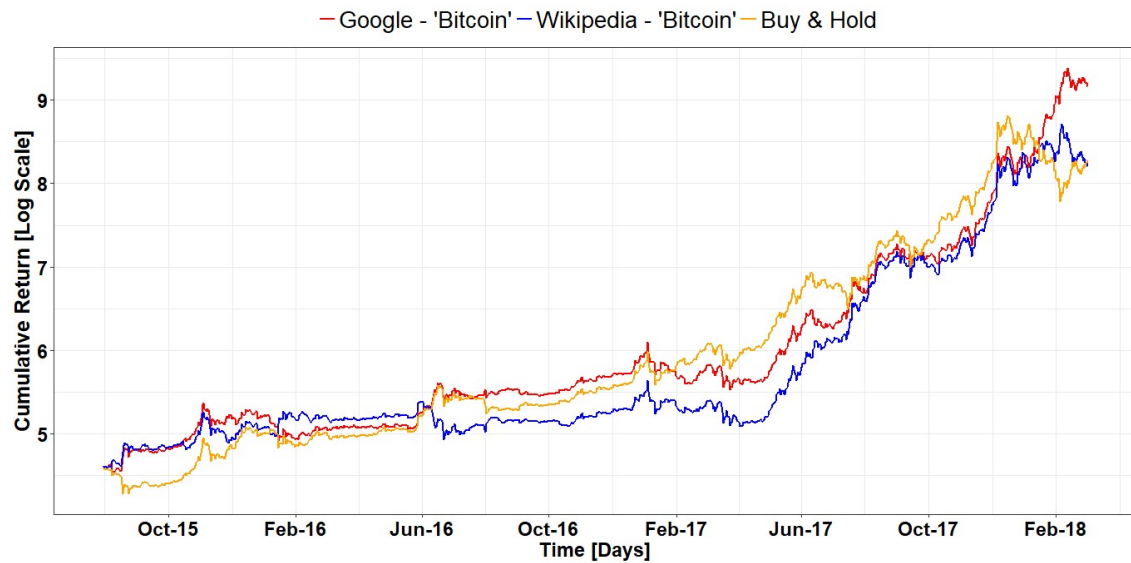
The Ann. Sharpe is calculated as the return per unit of risk (StdDev). The VaR at the 95% confidence interval (CI) reports the largest possible 1-day loss with a 5% probability of that loss occurring. ETL is defined as the average of all losses which are greater or equal than VaR at the 95% CI. The best performing search term is g.Bitcoin with a cumulative return of 9399.1% - significantly higher than the 3867.575% return yielded by the buy and hold strategy. The risk of the g.Bitcoin and w.Bitcoin strategy is also lower in terms of both the VaR and ETL. Although performing well, the w.Bitcoin strategy slightly under performs the buy and hold strategy.

Keyword	Cum. Ret [%]	Ann. Return[%]	Ann.Std Dev[%]	Ann.Sharpe	VaR [98%]	ETL [98%]
g.Bitcoin	9418.08	235.69	64.94	3.63	-8.35	-16.65
g.Cryptocurrency	82.94	17.42	65.51	0.27	-11.00	-21.50
g.Blockchain	771.80	77.82	65.36	1.19	-10.72	-21.60
g.Happiness	3344.01	156.20	65.15	2.40	-8.83	-19.08
g.Psychology	195.72	33.41	65.47	0.51	-10.12	-22.33
g.Big.Data	179.22	31.38	65.48	0.48	-11.22	-20.85
w.Bitcoin	3549.05	160.17	65.14	2.46	-8.10	-13.78
w.Cryptocurrency	128.44	24.56	65.49	0.38	-10.99	-21.44
w.Blockchain	379.22	51.67	65.43	0.79	-10.26	-22.29
w.Happiness	515.50	62.10	65.40	0.95	-9.95	-22.19
w.Psychology	373.38	51.18	65.43	0.78	-10.01	-22.24
w.Big.Data	993.32	88.85	65.33	1.36	-9.54	-21.54
Random	-0.23	-0.06	0.61	-0.10	-0.13	-0.15
Hold	3867.57	166.03	65.12	2.55	-9.91	-22.11

It is interesting to note that the only search term from the non-cryptocurrency related words to achieve high return is “Happiness”. Upon closer investigation, the search volume for happiness has no discernible trend. The median (trading signal) in this case would be close to zero (no change) each day which would result in primarily long positions being taken. This will result in the “Happiness” strategy closely resembling a buy and hold return.

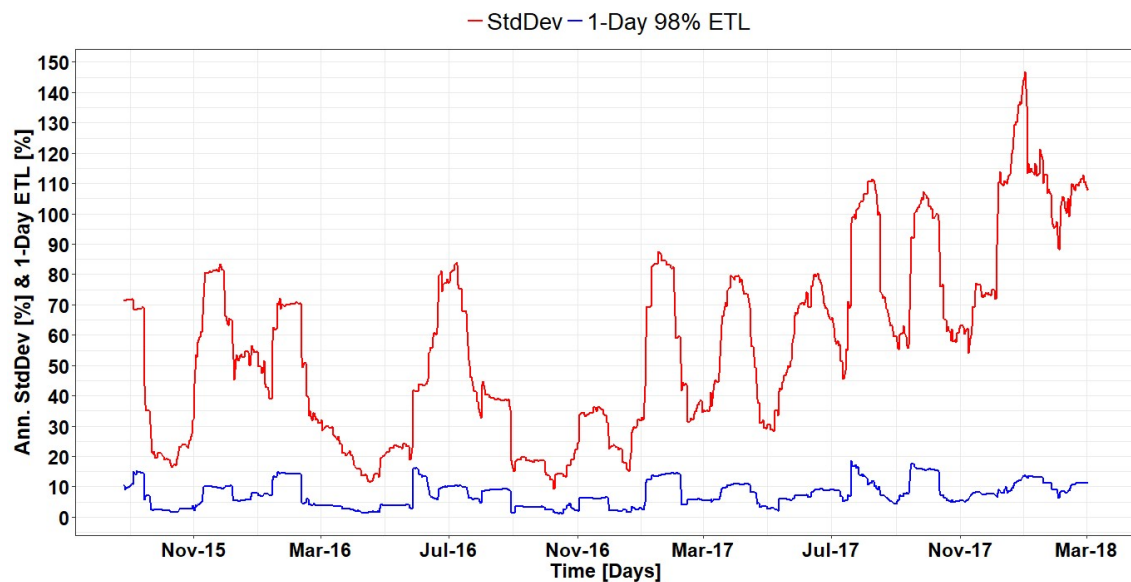
To put the absurdity of the above returns into perspective, the cumulative return of the Dow Jones Industrial Index since January 1990 (without adjusting for inflation) is 1815.713% with annualized return of 11.12%. However, it is imperative to note that the results obtained in this study are over a dramatically shorter time period and would have been close to impossible to achieve in practice. The likelihood of a trader employing a long/short cryptocurrency strategy using search term volume is also speculative.

Figure 3: Logarithmic cumulative returns of the Wikipedia and Google "Bitcoin" keyword strategy against the buy and hold strategy. The cumulative returns of all the strategies are closely linked. The "Bitcoin" Google Trends algorithm seems to time the BTC crash at the end of 2017 by taking short positions as search volume for BTC declined. Overall the Wikipedia strategy performs well, but is unable to avoid the severe Bitcoin price crash at the beginning of 2018



To give an insight to the volatility of the strategy returns, a rolling 28-day annualized standard deviation and expected tail loss with a 95% confidence interval plot for the g.Bitcoin keyword is displayed in Figure 3 below.

Figure 4: Rolling 28-day annualised StdDev and 95% confidence interval ETL. The standard deviation of returns has a clear upward trend which is highly unusual for a financial time series. StdDev seems to spike or cluster in waves – most notably at the end and middle of the year. ETL does not have a trend but closely follows the StdDev. Periods of high volatility seem to follow period of relatively low volatility. In other words, the volatility series is negatively autocorrelated.



3 Discussion

Predicting the price of Bitcoin and other cryptocurrencies is difficult. The high levels of volatility, speculation, insider trading and other issues such as exchange hacks create a challenging trading environment. Furthermore, BTC is still in its infancy and has seen a rapid rise in price and interest from larger industry players.

From the vast array of traditional technical analysis and algorithmic tools available to traders, could a simple search volume based algorithm employing Google and Wikipedia search term volumes be of use to traders to augment their existing strategies? The answer is mixed.

To uncover whether there exists a relationship between the price of BTC and related cryptocurrency search terms a correlation analysis was employed. By using a ‘stitching’ process described under the Methodology section, a daily time series of Google search volumes was derived from overlapping search volume history. Note that, unfortunately, the search volume history provided by Google is a random sample which is generated on a time basis which is unknown. For this reason, the validity of using Google search volume to generate a trading signal for a highly volatile

asset such as BTC is difficult because the signals are prone to change based on the random sample of search volume. Fortunately, the Wikipedia search volume is fixed and not subject to change. Although the random sampling of the Google search volume may seem trivial, with daily BTC returns often exceeding 10%, the generation of differing signals can have a profound effect on the final cumulative returns and overall profitability of the strategy. Although one could use a bootstrap approach to uncover the underlying distribution of Google search volumes, this would require intensive computing power.

The correlation between both the Wikipedia and Google search volumes for “Bitcoin” and the BTC price is striking. Although the correlation is time varying, the average correlation for the “Bitcoin” keyword and BTC price is 95%. Key words not associated with BTC have much lower levels of correlation (below 50%). Figure 1 clearly shows the co-movement of the price of BTC with the Wikipedia and Google search term for “Bitcoin”. It is interesting to note that the interest in BTC is one-sided. Search volume spikes as the price increases – and rapidly decreases after a price crash. This is essentially what allows the trading algorithm to perform so well. The behaviour of collective BTC investors is nicely captured by Google and Wikipedia – interest in price (speculation) is a primary driver of BTC returns and subsequent crashes.

The search term “Bitcoin”, yielded the lowest correlation to BTC price out of the cryptocurrency related keywords but returned the highest cumulative returns. Correlation alone does not determine the success of an algorithm. Due to the high volatility of search volume for BTC, a rolling 28-day median was computed and used as a trading rule described in the Methodology. By using the median, a large amount of noise can be eliminated. If the mean was used, the algorithm would be too slow to react to a change in search volume. Findings show support for using Wikipedia and Google search volume in trading BTC. All cryptocurrency search terms were highly profitable and the “Bitcoin” Google search term yields an astounding cumulative return. However, even though the results are overwhelmingly positive, the time period under review is short with the price of BTC experiencing unprecedented highs and then correcting to less than half of the all time high.

The implications of the findings are not simply that there is a correlation between search terms and BTC price with resulting profit from implementing a trading strategy. If used effectively, both Google and Wikipedia cryptocurrency search volume could effectively be used to investigate the overall state of the cryptocurrency markets and resulting bubbles and crashes. Prospective investors can time their entry and exit into the cryptocurrency markets based off collective human

sentiment measured by Google and Wikipedia.

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